

IPv4 & IPv6 & MAC Address

Internet Protocol

This guide covers the two major versions of the Internet Protocol — IPv4 and IPv6. You will learn their structure, addressing types, characteristics, advantages, and limitations.

Topic No.	Topic Name
Topic 1	IPv4 — Internet Protocol Version 4
Topic 2	IPv6 — Internet Protocol Version 6
Topic 3	Diffrence between IPv4 vs IPv6
Topic 4	MAC Address — Structure & Format
Topic 5	Parts of MAC Address (OUI & Device ID)

IPv4 (Internet Protocol Version 4)

1. Definition

IPv4 (Internet Protocol Version 4) is the most widely used protocol for identifying devices on a network. It assigns a unique IP address to each device so data can be sent and received over the internet.

2. Structure of IPv4 Address

- IPv4 is a 32-bit numeric address.
- It is written in dotted decimal notation.
- The address contains four numbers (octets) separated by dots.

Example: 192.168.0.1

Each octet ranges from 0 to 255.

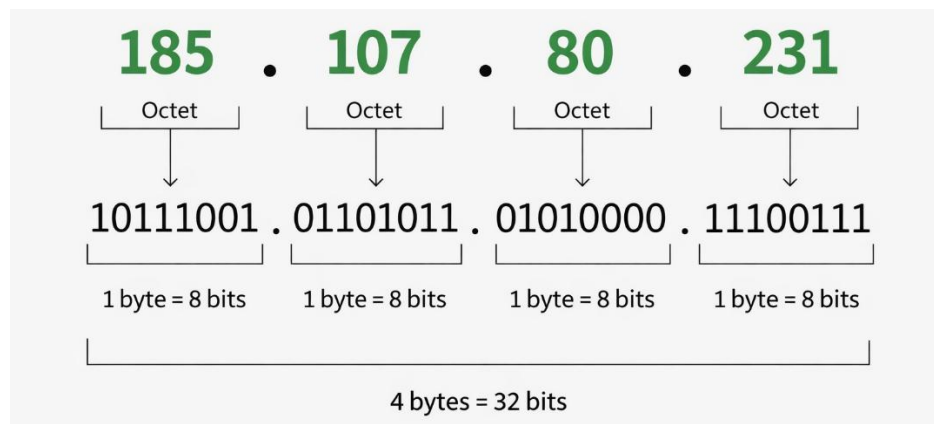


Figure 1: IPv4 address structure — 4 octets, each 8 bits, totalling 32 bits

3. Binary Representation

Computers store IPv4 addresses in binary form (32 bits).

Example: 185.107.80.231

Format	Octet 1	Octet 2	Octet 3	Octet 4
Decimal	185	107	80	231
Binary	10111001	01101011	01010000	11100111
NOTE	Binary IPv4: 10111001.01101011.01010000.11100111			

4. Types of IPv4 Addressing

1. Unicast: Communication between one sender and one receiver.

Example: Opening a website.

2. Broadcast: Communication from one device to all devices in a network.

Example: Sending a message to all devices in a LAN.

3. Multicast: Communication from one sender to multiple selected receivers.

Example: Video or audio streaming.

5. Characteristics of IPv4

- IPv4 uses 32-bit addressing.
- Address is written in numeric dotted format.
- Supports Unicast, Broadcast, and Multicast communication.
- Uses ARP (Address Resolution Protocol) to map IP addresses to MAC addresses.
- Supports VLSM (Variable Length Subnet Mask).

6. Advantages of IPv4

- Easy to connect multiple devices in a network.
- Supports encryption for privacy and security.
- Provides efficient communication and data transfer.

7. Limitations of IPv4

- Limited number of IP addresses.
- Complex network configuration.
- Difficult renumbering of addresses.

NOTE

Because of these limitations, IPv6 was introduced as the next generation Internet Protocol.

IPv6 (Internet Protocol Version 6)

- IP stands for Internet Protocol and v6 stands for Version Six (IPv6). It is the latest version of the Internet Protocol designed to replace IPv4. IPv6 is used to identify devices on a network and allows them to communicate over the internet.
- IPv6 uses 128-bit addresses, which provides a much larger number of unique IP addresses compared to IPv4. An IPv6 address is written using eight groups of hexadecimal numbers separated by colons.

1. Structure of IPv6 Address

- **Full form:** `2001:0db8:85a3:0000:0000:8a2e:0370:7334`
- Because IPv6 addresses are long, leading zeros can be removed and consecutive zero groups can be replaced with `::` for simplicity.
- **Short form:** `2001:db8:85a3::8a2e:370:7334`

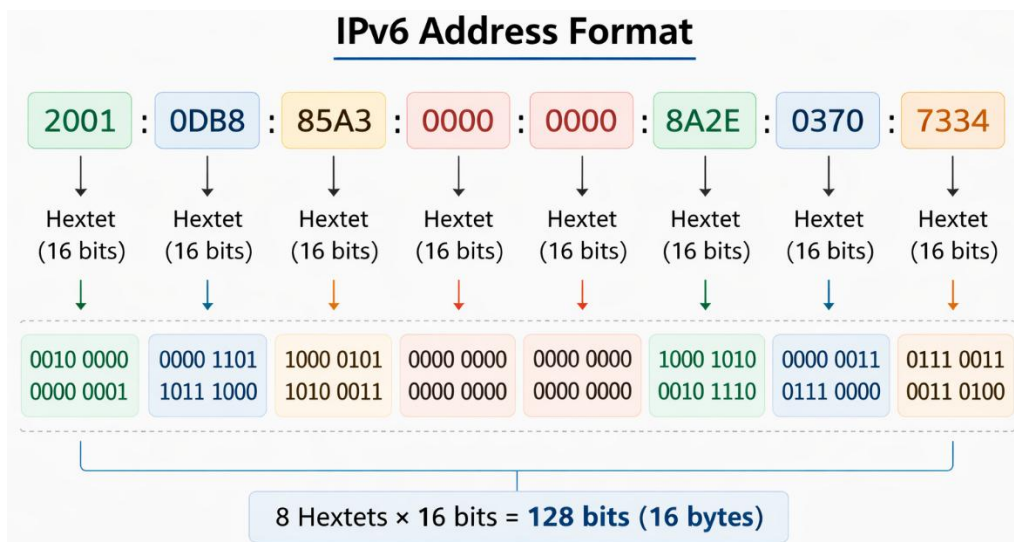


Figure 2: IPv6 address structure — 8 groups × 16 bits = 128 bits total

2. Hexadecimal Representation

IPv6 addresses are written in hexadecimal format, which uses numbers 0–9 and letters A–F.

Full address	2001:0db8:85a3:0000:0000:8a2e:0370:7334
Groups	8 groups, each containing 16 bits
Total bits	8 × 16 = 128 bits
Characters used	0–9 and A–F (hexadecimal)

3. Types of IPv6 Addressing

IPv6 supports three main types of addressing modes:

- 1. Unicast Addressing Mode:** This addressing mode is used to send data from one sender to one receiver.
Example: Accessing a specific website or server.
- 2. Multicast Addressing Mode:** This addressing mode is used to send data to multiple devices in a group.
Example: Video streaming to multiple users.
- 3. Anycast Addressing Mode:** This addressing mode sends data to the nearest device in a group of devices.
Example: Content Delivery Networks (CDN).

4. Advantages of IPv6

- Provides a very large number of IP addresses.
- Improves network efficiency and routing performance.

- Supports better security with IPsec.

5. Limitations of IPv6

- Requires upgrading network hardware and software.
- Transition from IPv4 to IPv6 can be complex.
- Some networks and systems still rely on IPv4.
- Implementation and migration may increase cost.

IPv4 vs IPv6 — Comparison Table



Figure 3: IPv4 vs IPv6 side-by-side visual comparison

Feature	IPv4	IPv6
Address Bits	32-bit	128-bit
Notation	Dotted Decimal e.g. 192.168.0.1	Hexadecimal e.g. 2001:db8::1
Address Types	Unicast, Broadcast, Multicast	Unicast, Multicast, Anycast
Security	Optional (external IPsec)	Built-in IPsec
Auto Config	Manual or DHCP	SLAAC (automatic)
Header Size	20–60 bytes (variable)	40 bytes (fixed)
Broadcast	Supported	Not supported (uses Multicast)

NOTE	IPv4 and IPv6 can run side by side during transition using a technique called Dual Stack , a device supports both protocols simultaneously.
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MAC Address (Media Access Control Address)

- A MAC Address (Media Access Control Address) is a unique hardware identifier assigned to a network interface card (NIC) of a device. It is used to identify devices within a local network and allows communication between devices at the data link layer of the OSI model.
- The MAC address is assigned by the manufacturer of the network device and is stored in the hardware of the network interface card.

Structure of MAC Address

- A MAC Address (Media Access Control Address) is a 48-bit unique hardware address assigned to a network interface card (NIC) of a device.
- It is used to identify devices within a local area network (LAN) and works at the Data Link Layer (Layer 2) of the OSI Model.

Format of MAC Address

- A MAC address is written in hexadecimal format and consists of six groups of two hexadecimal digits.

Format	XX : XX : XX : XX : XX : XX
Each pair	8 bits (1 byte)
Total length	48 bits (6 bytes)
Notation	Hexadecimal (0–9 and A–F)

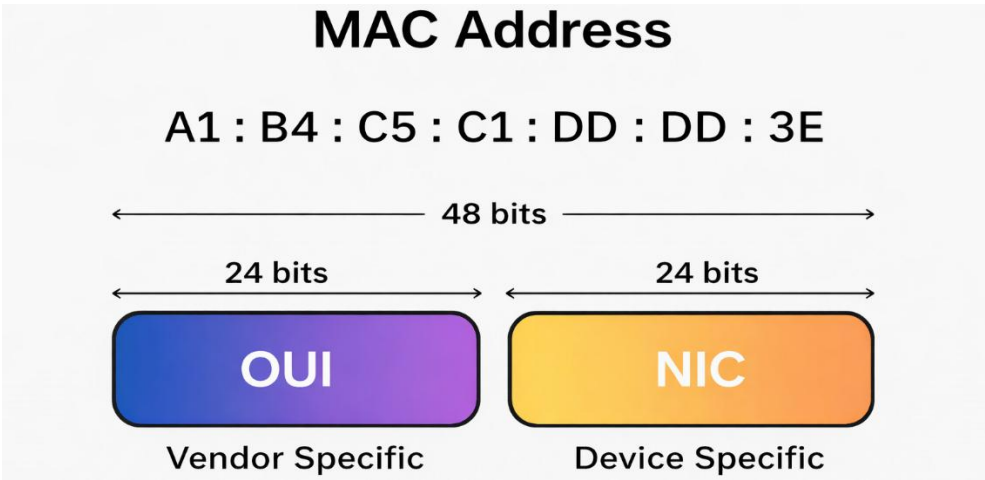


Figure 4: MAC address structure — OUI (first 24 bits) and Device Identifier (last 24 bits)

Parts of MAC Address

A MAC address is divided into two main parts:

1. OUI (Organizationally Unique Identifier)

- The first 24 bits (first three bytes) of the MAC address.
- It identifies the manufacturer or organization that produced the network device.
- This portion is assigned by the IEEE (Institute of Electrical and Electronics Engineers).

2. Device Identifier

- The last 24 bits (last three bytes) of the MAC address.
- It is assigned by the manufacturer.
- This part uniquely identifies the specific device.

Key Points of MAC Address Structure

- **Total length:** 48 bits (6 bytes)
- **Written in** hexadecimal format
- **Divided into two parts:** OUI and Device Identifier
- **Used for** device identification in local networks
- **Works at** Layer 2 of the OSI Model

KEY

Every device manufactured has a globally unique MAC address. No two devices should ever have the same MAC address - it is like a permanent digital fingerprint for your network card.